

Zero belongs to the reciprocal Hurwitz spectrum

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Conjecture 00000008347

Result. The proposed closed interval $[1/\sqrt{5}, 1/2]$ omits the value zero attained by every Liouville number.

Convention and scope. For a real number t , let $\|t\|_{\mathbb{Z}} = \min_{p \in \mathbb{Z}} |t - p|$. The reciprocal Lagrange (or Hurwitz approximation) constant is

$$c(\alpha) = \frac{1}{\nu(\alpha)} = \liminf_{q \rightarrow \infty} q \|q\alpha\|_{\mathbb{Z}}.$$

This is the usual reciprocal-constant convention; see [1]. Infinite $\nu(\alpha)$ is allowed, with reciprocal zero. Both source languages specify *all irrationals*, with no bounded-partial-quotient restriction or exclusion of zero. Set

$$\mathcal{S} = \{c(\alpha) : \alpha \in \mathbb{R} \setminus \mathbb{Q}\}.$$

We refute the claimed equality $\overline{\mathcal{S}} = [1/\sqrt{5}, 1/2]$.

The actual approximation errors. Let $L = \sum_{n=0}^{\infty} 10^{-n!}$. This is an irrational Liouville number. Its rational approximations with exponent three occur at arbitrarily large denominators: there are unbounded positive integers q and integers p such that

$$0 < |L - p/q| < q^{-3}.$$

Nearest-integer minimization then gives

$$0 \leq q \|qL\|_{\mathbb{Z}} \leq q |qL - p| = q^2 |L - p/q| < q^{-1}.$$

For every $\varepsilon > 0$, such q can be chosen arbitrarily large with $q^{-1} < \varepsilon$. Thus the sequence of nonnegative errors has limit inferior zero, so $c(L) = 0$. This argument also proves the same assertion for every Liouville number, directly from its full rational approximation property.

Contradiction in the actual spectrum. Since L is irrational, $0 = c(L) \in \mathcal{S} \subseteq \overline{\mathcal{S}}$. But $1/\sqrt{5} > 0$, so $0 \notin [1/\sqrt{5}, 1/2]$. The claimed closed-interval spectrum is therefore false. No conclusion about the conjecture's separate realization dimensions, golden-ratio uniqueness, or gap claims is needed.

Formal verification. Lean uses the actual factorial infinite series, Mathlib's full Liouville approximation predicate and its arbitrarily-large-denominator theorem. The error is $q|q\alpha - \text{round}(q\alpha)|$, and Mathlib's verified nearest-integer inequality supplies the minimization bridge. The proof establishes both bounds needed for the genuine real `Filter.liminf`, proves its value is zero, and establishes membership in the actual set and its topological closure. The final theorem `Counterexample.counterexample` is the negation of the displayed set equality. The reciprocal convention is stated explicitly; no numerical approximation or assumed spectrum membership is used.

References

- [1] D. Gayfulin, *Attainable numbers and the Lagrange spectrum*, Acta Arithmetica **179** (2017), 185–199. The reciprocal-constant definition is stated in the abstract.
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